

Solution Architecture

Date: 30 September 2026

Team ID: SWTID-2026-4040

Project Name: ComicCraft — AI Comic Story Creator Using Gemini Models

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Team Members: Sandhiya D, Rithikaa B, Anitha R, Anbuselvi

Team Size: 5

Maximum Marks: 5

High-Level Architecture

[Presentation Layer: Jinja2 HTML]



[FastAPI Application / Routes]



[AI Orchestration]



Gemini Flash Gemini Pro Stable Diffusion



Outline Story Text Panel Images



[Layout Builder]



[Comic Preview / Export Flow]

Component Name	Description / Role	Technology
Presentation Layer	Collects inputs and renders output pages.	HTML, Jinja2
API / Application	Defines routes and orchestrates services.	FastAPI, Uvicorn
Outline Generator	Creates a five-panel outline and image prompts.	Gemini 1.5 Flash
Story Generator	Expands panel outline into narration/dialogue.	Gemini 1.5 Pro
Image Generator	Creates panel artwork from prompts.	Diffusers, Stable Diffusion, PyTorch
Layout Builder	Combines panel metadata, story and image paths.	Python
Export Flow	Provides route/page for export completion; referenced exporter is not included in upload.	FPDF referenced in requirements; exporter module absent